

Today's Agenda

1) Transformers

↳ Attention is All You Need

Seq 2 Seq

Self-Attention

Context

Not on Proximity.

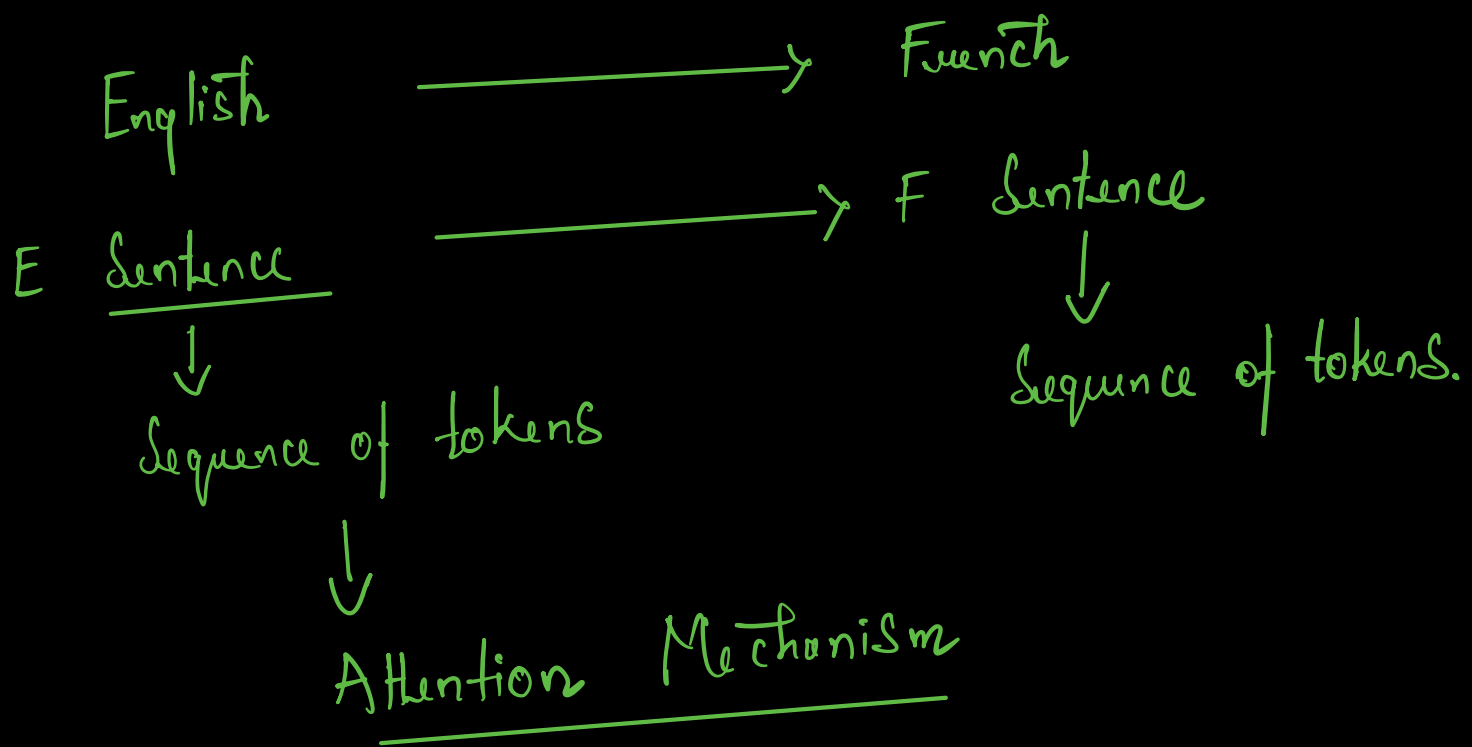
My name is Paul and I am going London

Sequence 2 Sequence

Encoder Decoder

RNN :- Many to Many

Machine Translation, Text Summarization, Text Generation

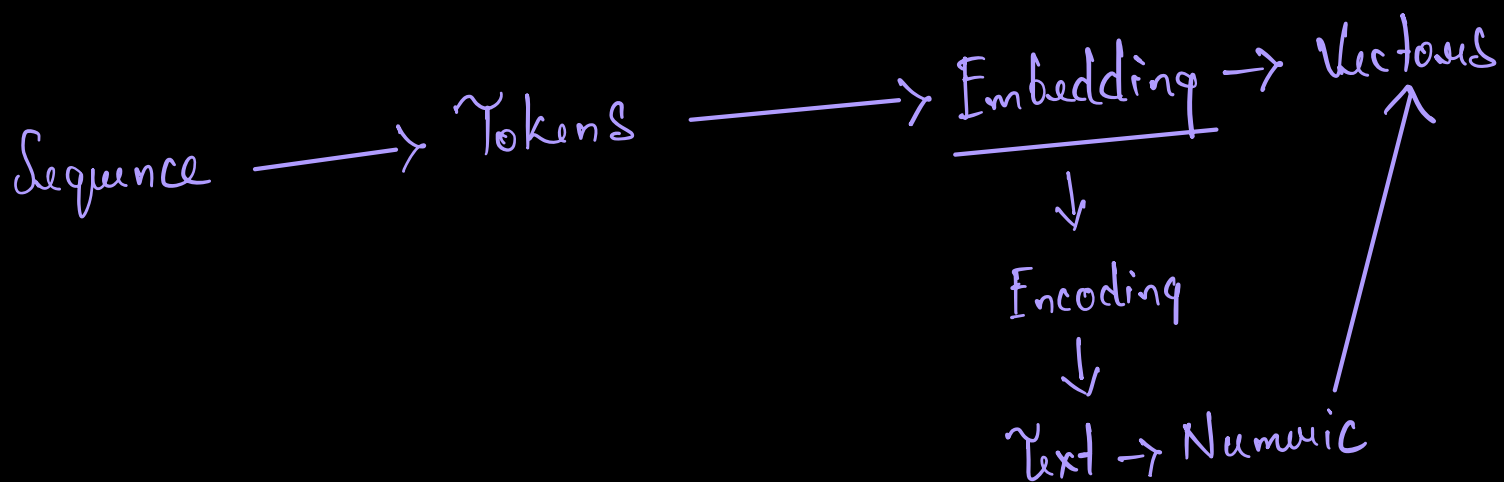


2022 LLM :- Large Language Models

Parameters :- 10 Billion

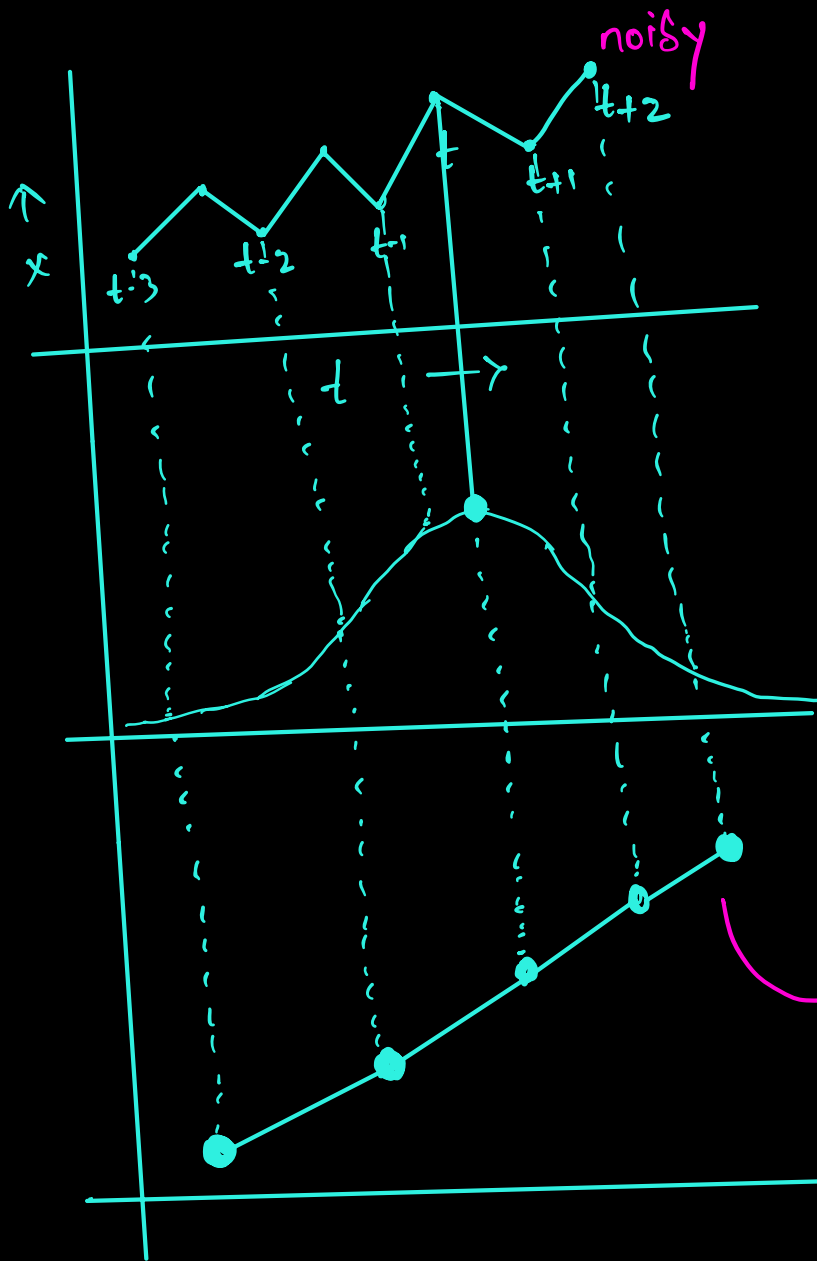
2018 SLM :- Small Language Models

500M ~ 10 Billion



ATTENTION MECHANISMS

Time Series Data



EWA
Exponential Weighted Avg

Reweighing Scheme

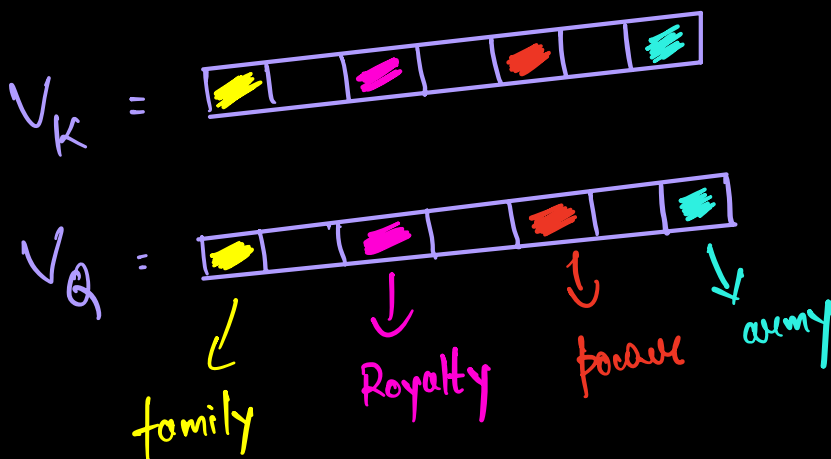
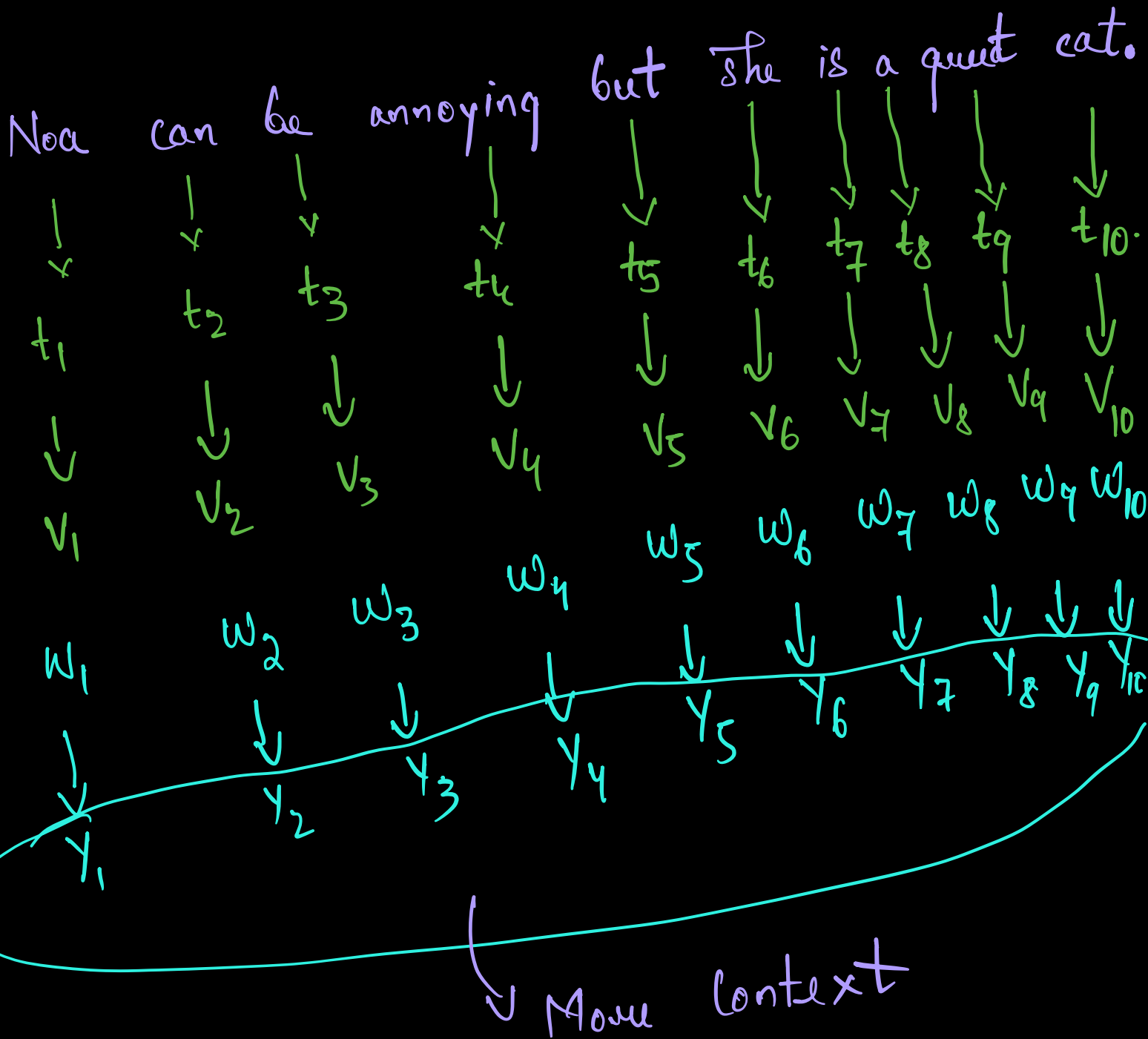
$$W_i = [w_0, w_1, w_2, w_3]$$

1) Filter data that is far away.

2) Amplify if something is close.

Useful data points gave context with each another.

Focus on the Reweighing Scheme.



Similar Words

Son
daughter
army
Kingdom
...

Reweighting Scheme / Plan

$$\begin{aligned} W_{kQ} &= V_Q \cdot V_k \\ &= V_k - V_Q \end{aligned}$$

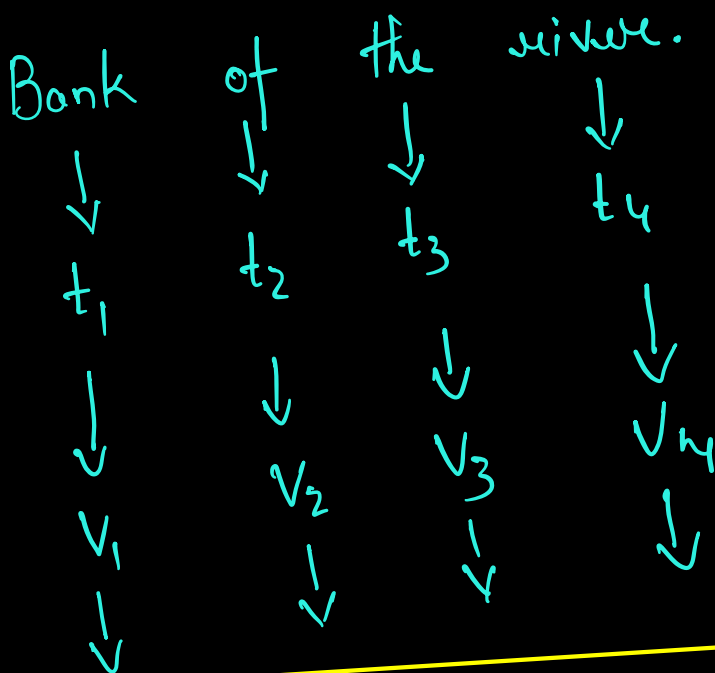
→ Explode

country
land

Dog, cat, computer
Swimming

Today's Agenda

SELF ATTENTION



REWEIGHT

$\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 Y_1 Y_2 Y_3 $Y_4 \leftarrow$ Better contextualized vectors

$$V_1 V_1 = W_{11}$$

$$V_1 V_2 = W_{12}$$

$$V_1 V_3 = W_{13}$$

$$V_1 V_4 = W_{14}$$

$\longrightarrow$
 Normal

$$W_{11}$$

$$W_{12}$$

$$W_{13}$$

$$W_{14}$$

$\longrightarrow$ Sum to 1

$$W_{11} V_1 + W_{12} V_2 + W_{13} V_3 + W_{14} V_4 = Y_1$$

Reweighing all vectors towards V_1

$$W_{21} V_1 + W_{22} V_2 + W_{23} V_3 + W_{24} V_4 = Y_2$$

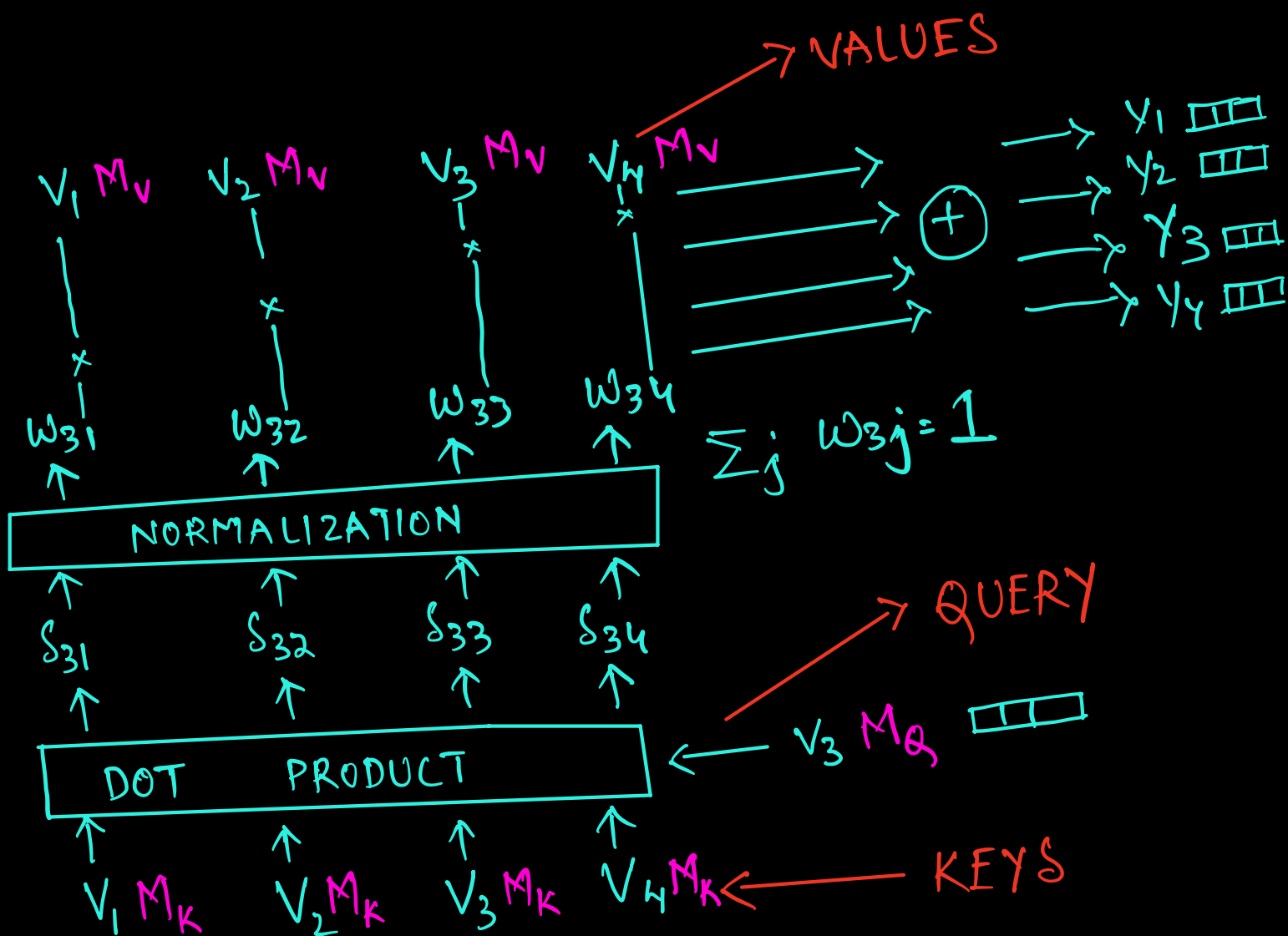
$$W_{31} V_1 + W_{32} V_2 + W_{33} V_3 + W_{34} V_4 = Y_3$$

$$W_{41} V_1 + W_{42} V_2 + W_{43} V_3 + W_{44} V_4 = Y_4$$

Benefits of the Reweighting Scheme

- ① I have not trained any weights
- ② Order has no influence
- ③ Proximity has no influence
- ④ Shape Independent

SELF - ATTENTION

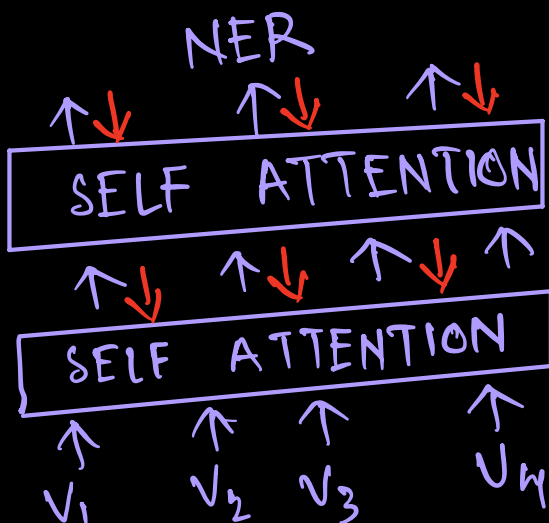
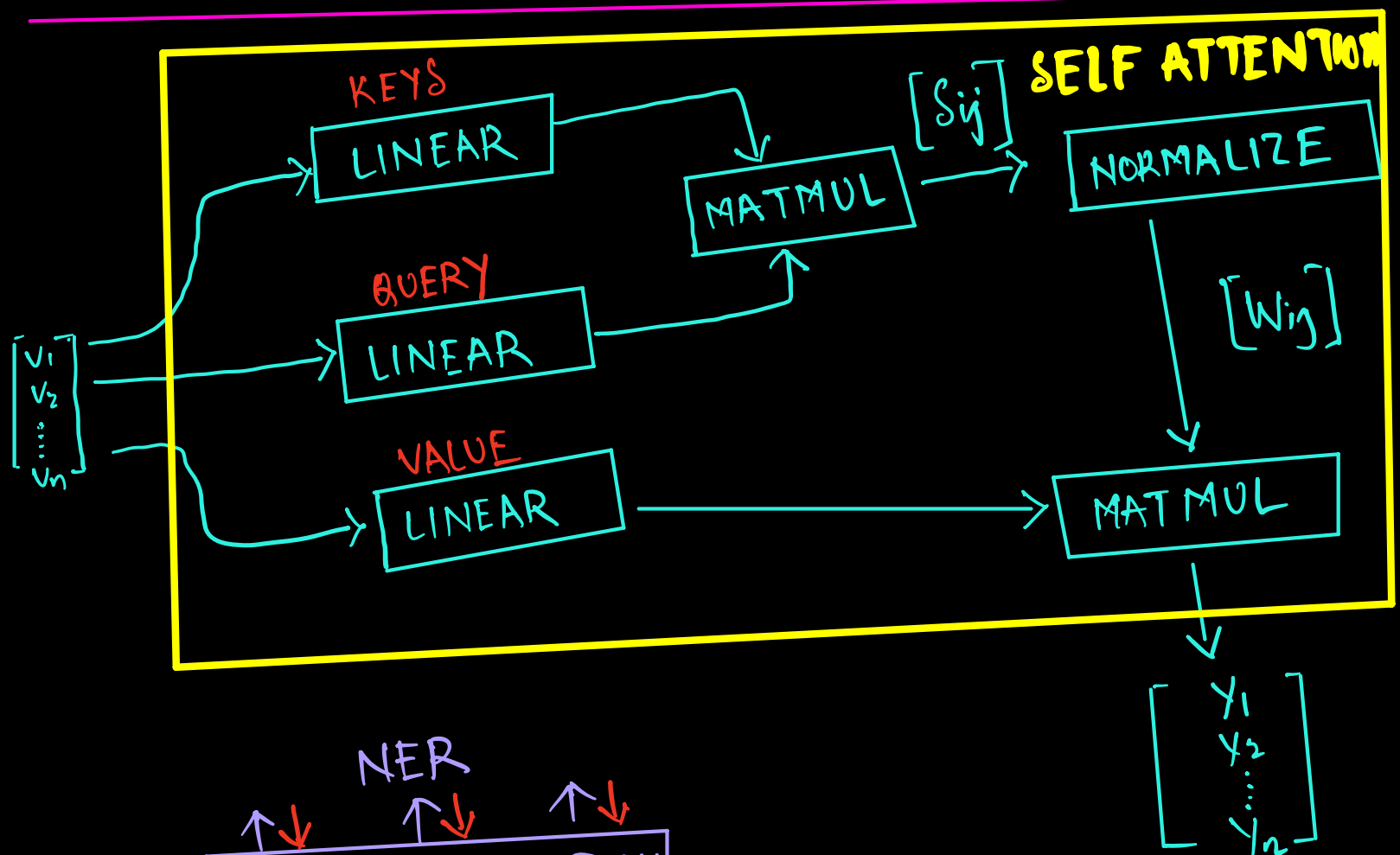




$$V_i = (1 \times K)$$

$$M = (K \times K)$$

$$V_i \cdot M = (1 \times K)$$



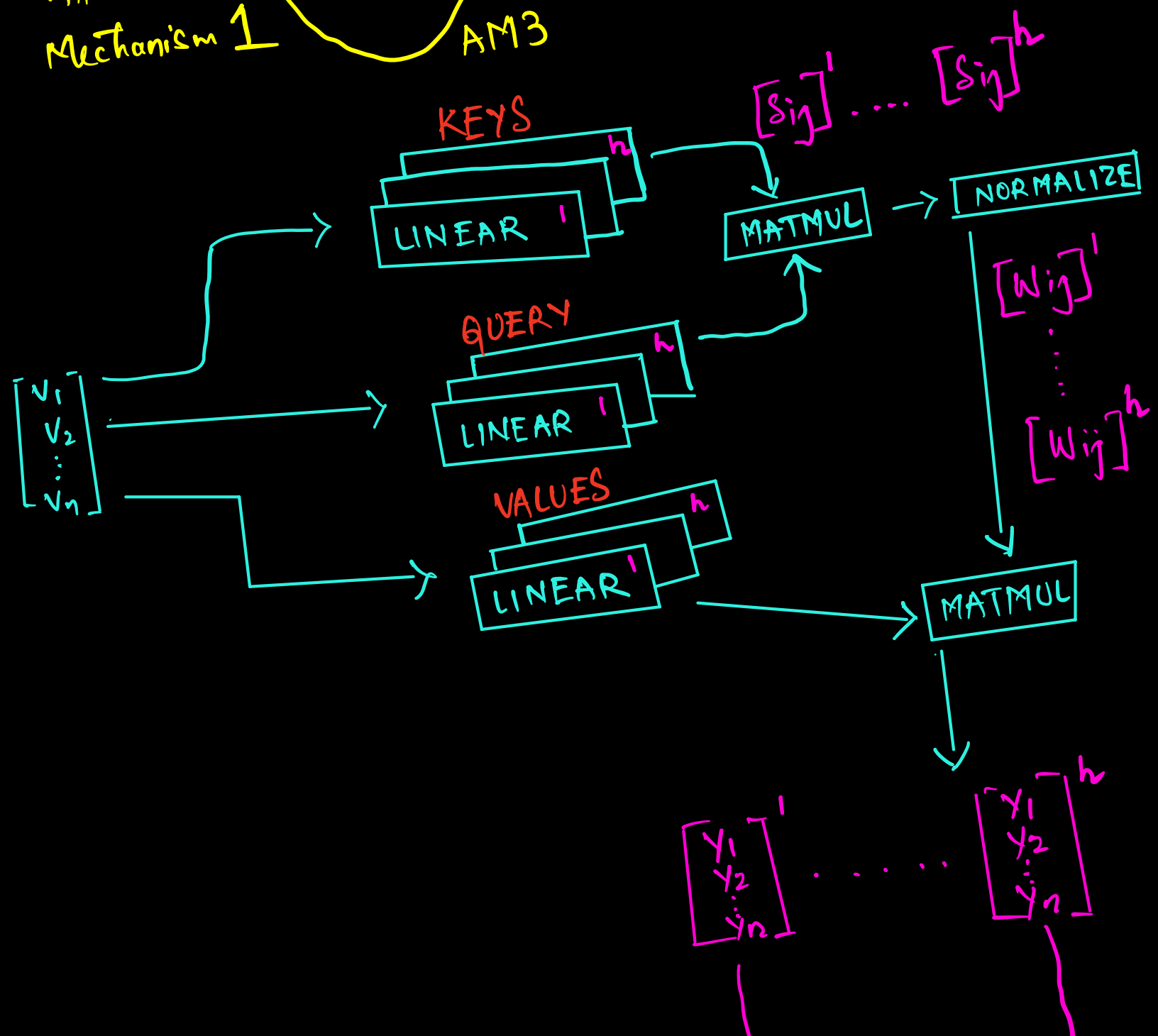
Do we have enough attention?

I gave my food to Charlie.

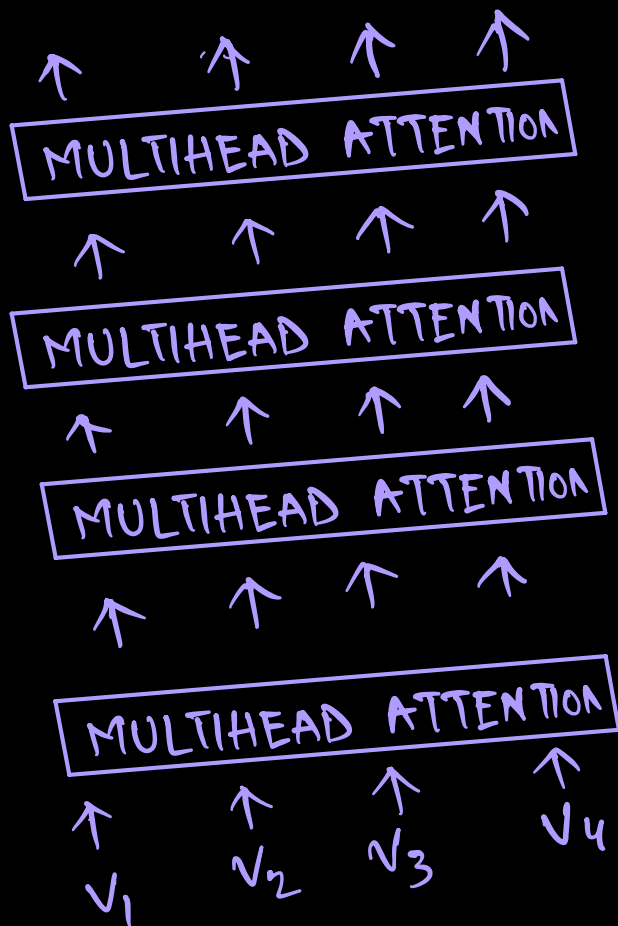
Attention Mechanism 1

AM2

AM3

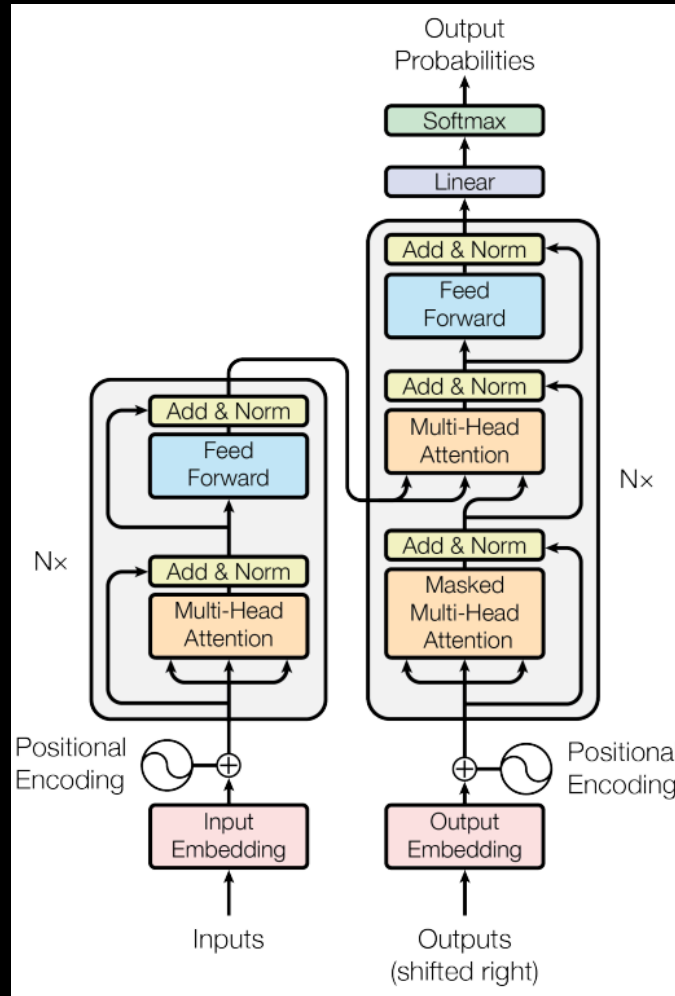


MULTIHEAD ATTENTION



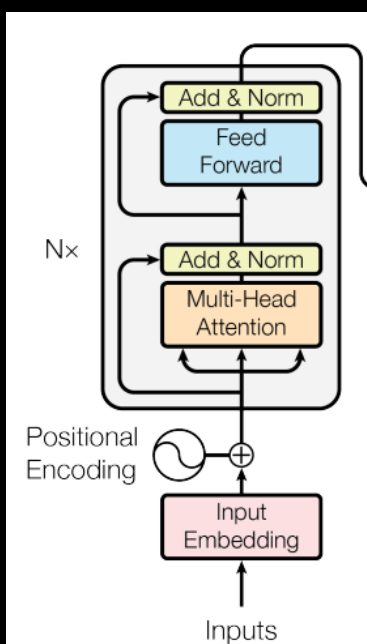
CONCAT + DENSE

$$\begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_n \end{bmatrix}^*$$



Encoder

Order has influence
Positions



P.E $\rightarrow$ Act Embedding

$$\begin{array}{|c|c|c|c|c|c|} \hline & & & & & \\ \hline \end{array} + \begin{array}{|c|c|c|c|c|c|} \hline & & & & & \\ \hline \end{array}$$

$$P(E, V)$$

$$\begin{array}{|c|c|c|c|c|c|} \hline & & & & & \\ \hline \end{array}$$

$$(\sin, \cos)$$

$$\Downarrow$$
Final Encoding

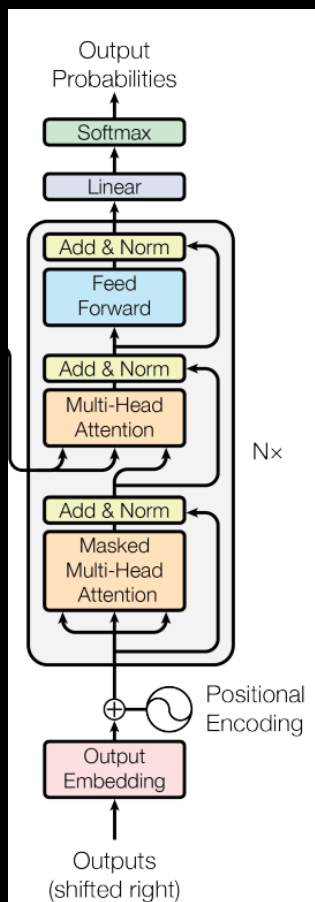
Vanishing gradient

Feed Forward Network

Linear
dropout
linear

activation function

Non linearity



DECODER

Mask

↳ Hide

Standard M.H.A

My cat sat on the floor.

MASK · M.H.A

My cat ○ ○ ○ ○

itself and past

* Nothing in future *

MASK · M·H·A

A·M

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) V$$

M·A·M

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) + \underset{\substack{\downarrow \\ \text{Mask Matrix}}}{M}\right) V$$

Mask

Casual Mask

mask = 4 × 4

$$= \begin{bmatrix} 0 & -\text{inf} & -\text{inf} & -\text{inf} \\ 0 & 0 & -\text{inf} & -\text{inf} \\ 0 & 0 & 0 & -\text{inf} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Token 1

Token 1 & 2

$$\text{softmax}(-\text{inf}) \approx 0$$

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

→ Even

→ Odd

i = dimension index

512

d_{model} = Embedding length

pos = position of token in a sequence.

Why ?

Periodicity
Constrained values.